

SMART ENERGY MONITORING AND POWER CONSUMPTION COST ANALYSIS SYSTEM USING ESP32 AND IoT

PROJECT REPORT

Submitted by

Khanderao Wani

Final Year Project Report

ABSTRACT

Electricity is an essential resource in homes, industries, educational institutions and commercial buildings. Conventional electricity monitoring systems often provide only total consumption information. The Smart Energy Monitoring and Power Consumption Cost Analysis System Using ESP32 and IoT is designed to monitor and visualize voltage, current, power, energy consumption, operating status and estimated electricity cost through an interactive web-based dashboard.

The system follows an ESP32-oriented IoT architecture. Electrical values are processed to calculate instantaneous power, accumulated energy and estimated cost. The dashboard provides live numerical values, graphical visualization, monitoring history, activity logs and operating status. The project also demonstrates NORMAL, WARNING and FAULT/OVERCURRENT conditions with LED-based status indication.

Start Simulation and Stop Simulation controls allow the monitoring logic to be activated or made idle. When stopped, simulation updates, graphs, energy and cost calculations, history updates and uptime timing are stopped.

1. INTRODUCTION

Energy monitoring is increasingly important because efficient electricity utilization requires awareness of present operating conditions. Conventional meters may not provide an interactive view of voltage, current, power, energy usage, estimated cost or abnormal conditions. The proposed system addresses this by processing electrical data and presenting it on a user-friendly web dashboard.

2. PROBLEM STATEMENT

The major problems addressed are difficulty in continuously monitoring electrical parameters, lack of easy visualization of voltage, current and power, limited awareness of real-time energy consumption, difficulty in estimating electricity cost, lack of immediate indication of abnormal current conditions and limited access to graphical monitoring data.

3. OBJECTIVES

- Monitor voltage and current.
- Calculate electrical power.
- Calculate energy consumption.
- Estimate electricity cost.
- Display live electrical data.
- Provide graphical visualization.
- Detect NORMAL, WARNING and FAULT/OVERCURRENT conditions.
- Provide Start, Stop and Reset control.

4. SCOPE OF THE PROJECT

The scope includes smart energy monitoring, voltage and current monitoring, power calculation, energy consumption analysis, electricity cost estimation, a web-based dashboard, graphical data visualization, fault and warning indication, ESP32-based IoT architecture and simulation-based implementation.

5. SYSTEM OVERVIEW

The project consists of data acquisition, processing and visualization. Voltage and current values are obtained from simulation logic or may later be supplied by physical sensors. The ESP32 acts as the central controller concept, processing values and supporting power calculation and status detection. The dashboard displays voltage, current, power, energy, electricity cost, system status, uptime, graphs, monitoring history and activity logs.

6. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware or future implementation components include ESP32, voltage sensing, a current sensor, LED, connecting wires and a computer. Software technologies include Wokwi for ESP32 simulation, HTML for webpage structure, CSS for design, JavaScript for simulation and dynamic calculations, a web browser for dashboard execution and ESP-IDF/C for ESP32 programming.

7. ESP32

The ESP32 is used as the central controller concept because it supports GPIO interfacing, ADC operation, processing capability and Wi-Fi connectivity suitable for IoT applications. It processes electrical information and supports the smart monitoring architecture.

8. WORKING PRINCIPLE

Step 1: The user starts the simulation.

Step 2: The system obtains or generates voltage and current data. An example condition is Voltage = 230 V and Current = 2 A.

Step 3: Power is calculated using $P = V \times I$. Therefore, $230 \times 2 = 460$ W.

Step 4: Energy is calculated using $\text{Energy (kWh)} = \text{Power (W)} \times \text{Time (hours)} / 1000$.

Step 5: Electricity cost is estimated using $\text{Cost} = \text{Energy (kWh)} \times \text{Tariff}$.

Step 6: The dashboard updates live values, graphs, status, history and activity information while the simulation is active.

9. SYSTEM STATUS MONITORING

NORMAL indicates that the monitored condition is within the configured normal range. WARNING indicates that a predefined warning level has been reached. FAULT or OVERCURRENT indicates that the monitored current exceeds the configured safe threshold.

10. LED STATUS INDICATION

An LED provides a visual indication of system status or fault conditions. The ESP32 GPIO output controls the LED through the project circuit, making the demonstration easier to understand.

11. WEB-BASED DASHBOARD

The dashboard is the main user interface and displays the name Khanderao Wani. It includes live voltage, current, instantaneous power, accumulated energy, estimated electricity cost, system status, uptime, current and power graphs, monitoring history, activity logs and Start Simulation, Stop Simulation and Reset controls.

12. START SIMULATION FUNCTION

Start Simulation activates monitoring or data generation, updates voltage and current, calculates power, accumulated energy and electricity cost, updates graphs and monitoring history, updates activity logs and starts the uptime timer.

13. STOP SIMULATION FUNCTION

Stop Simulation stops data generation or monitoring updates, graph updates, energy calculation, cost calculation, monitoring history updates, simulation-driven activity updates and the uptime timer. The monitoring logic therefore remains idle except for normal browser activity.

14. RESET SYSTEM FUNCTION

The reset function returns the dashboard to its initial monitoring state and can clear or reinitialize energy, cost, graph data, monitoring history, activity logs, uptime and status.

15. BLOCK DIAGRAM

Voltage / Current Sensor or Simulation → ESP32 → Data Processing → Power Calculation / Energy Calculation / Cost Calculation / Status Detection → Web Dashboard. The dashboard provides live data, graphs, logs and energy and cost analysis. An LED can provide visual status indication.

16. MATHEMATICAL CALCULATIONS

Voltage is represented by V in volts. Current is represented by I in amperes. Power is $P = V \times I$ in watts. Energy is $E(\text{kWh}) = P(\text{W}) \times t(\text{hours}) / 1000$. Electricity cost is $\text{Cost} = E(\text{kWh}) \times \text{Tariff}$.

Example: Voltage = 230.0 V, Current = 2.0 A, Power = 460.0 W and Status = NORMAL.

17. RESULTS AND OBSERVATIONS

The project demonstrates voltage and current monitoring, automatic power calculation, energy tracking, electricity cost estimation, system status monitoring, warning and fault demonstration, LED-based indication, graphical visualization, monitoring history, activity logs and Start, Stop and Reset controls. The uptime timer and simulation-driven updates stop when Stop Simulation is selected.

18. ADVANTAGES

The system provides centralized monitoring, real-time numerical and graphical visualization, energy awareness, estimated cost information, abnormal-condition demonstration, user-controlled operation, a web-based interface and an expandable architecture for future IoT and sensor integration.

19. LIMITATIONS

The present demonstration is simulation and web-dashboard based. Future physical accuracy depends on sensor selection and calibration. Electricity cost is an estimate based on the configured tariff. Real hardware implementation requires safe electrical isolation and interfacing, while cloud deployment would require secure communication and data storage.

20. FUTURE SCOPE

Future improvements can include real voltage and current sensors, cloud database integration, remote IoT monitoring, a mobile application, multiple appliance monitoring, notification alerts, historical energy analysis, consumption prediction and automatic load control.

21. APPLICATIONS

Applications include smart homes, educational laboratories, industrial monitoring, offices, commercial buildings, energy management systems, IoT research projects and electrical engineering laboratories.

22. CONCLUSION

The Smart Energy Monitoring and Power Consumption Cost Analysis System Using ESP32 and IoT demonstrates an integrated approach for observing and analyzing electrical parameters. The project processes voltage and current information to calculate power, energy consumption and estimated electricity cost. These values are presented through an interactive web-based dashboard with numerical displays, graphs, history, activity logs and system status information.

The inclusion of NORMAL, WARNING and FAULT conditions demonstrates basic abnormal-condition monitoring, while LED indication provides a simple physical representation of system status. Start and Stop Simulation controls provide user control over active processing. When stopped, simulation updates, graph updates, energy and cost calculations and the uptime timer stop.

The project provides a foundation for future expansion into a real sensor-based smart energy monitoring system.

23. REFERENCES

1. ESP32 Technical Documentation. 2. Wokwi ESP32 Simulation Documentation. 3. ESP-IDF Programming Documentation. 4. HTML, CSS and JavaScript documentation. 5. Electrical energy measurement and power monitoring textbooks. 6. IoT-based smart energy monitoring research literature.